

SSEE as a Working Effective Field Theory: K-essence Action, Algebraic Coupling, and Bellini-Sawicki Classification

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Abstract

We present a working effective field theory (EFT) formulation of SSEE at the background and linear-perturbation level. The fundamental action is a k -essence scalar field with a non-canonical kinetic term $K(X) = X/\text{KAL}_0 + X^2/M^4$ and an exponential potential $V(\phi) = V_0 e^{-\alpha\phi}$, minimally coupled to gravity and conformally coupled to dark matter. All parameters are fixed algebraically by the golden ratio φ and π with zero free parameters: the structural viscosity $\text{KAL}_0 = \beta + \pi \approx 5.521$, the tracker exponent $\alpha = \lambda/\sqrt{\text{KAL}_0} \approx 0.2948$, and the dark-sector coupling constant $\beta_c = -\mathcal{A} \equiv -(3\varphi + \pi)/2 \approx -3.998$. Within the model sector studied here, we derive analytic conditions under which the Null Energy Condition is satisfied and $c_s^2 \in (1/3, 1)$, corresponding to ghost-free and gradient-stable perturbations in that regime. The phantom crossing $w_{\text{eff}} < -1$ observed by DESI DR2 is reproduced via the covariant energy-transfer four-vector Q^ν , without any NEC violation in the fundamental sector. We derive the working background and linear perturbation equations, construct the Bellini-Sawicki classification ($\alpha_T = \alpha_B = \alpha_M = 0$, $c_s^2 \neq 1$, $\alpha_K \neq 0$), and provide all inputs required for numerical implementation in `HI_CLASS/EFTCAMB`. The numerical implementation reproduces $\Omega_\phi(a=1) = 0.839950$ to 10^{-12} under the adopted initial conditions and yields $c_s^2 \in [0.60, 0.95]$ over $a \in [0.1, 1]$.

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1 Introduction

The accelerated expansion of the universe (Perlmutter et al., 1999; Riess et al., 1998) has motivated a broad landscape of dynamical dark energy models beyond the cosmological constant (Copeland, Sami & Tsujikawa, 2006; Weinberg et al., 2013; Clifton et al., 2012; Joyce et al., 2015). Within this landscape the Chevallier–Polarski–Linder (CPL) parametrization (Chevallier & Polarski, 2001; Linder, 2003) provides the standard observational proxy; its parameters (w_0, w_a) are now constrained by DESI DR2 at 3.1σ from Λ CDM (Abdul-Karim et al., 2025).

The Structural Self-Energy Expansion (SSEE) is a cosmological dark energy model in which the equation-of-state parameters $w_0 = -T_r/M_v \approx -0.840$ and $w_a = -P_{sc}/I_g \approx -0.670$ emerge algebraically from the golden ratio $\varphi = (1 + \sqrt{5})/2$ and π , with no free parameters (Almeida Vallejo, 2025a,b). The algebraic point $(w_0, w_a) = (-0.840, -0.670)$ falls within the DESI 1σ contour at 0.24σ (Pantheon+) (Almeida Vallejo, 2025b).

Papers 1–6 of this series established the phenomenological description, Bayesian validation (MCMC vs DESI+Planck), CMB confrontation, algebraic H_0 derivation, interacting-sector perturbation theory, and a two-sector ϕ -DM model (Almeida Vallejo, 2026b) that resolves the S_8 lensing tension (Asgari et al., 2021; Dark Energy Survey Collaboration, 2022) from 3.5σ (single-sector ceiling, KiDS-1000, Paper 5) to 0.04σ (two-sector ϕ -DM), while the $f\sigma_8$ growth-rate tension is already resolved at the single-sector level (0.70σ , statistically identical to Λ CDM 0.73σ) (Almeida Vallejo, 2026a). A persistent methodological concern, however, is that a model specified solely by (w_0, w_a, Ω_m) remains *phenomenological*: it parametrizes dark energy without providing a Lagrangian from which the equations of motion, stability conditions, and perturbation spectrum can be derived. Foundational EFT frameworks for dark energy (Cheung et al., 2008) address this gap for generic scalar fields; SSEE goes further by fixing all EFT coefficients algebraically from φ and π .

This paper closes that gap. We present the working EFT architecture: the fundamental action (§2), structural matching of the potential (§3), analytic NEC proof and scalar sound speed (§4), covariant dark-sector coupling with full algebraic closure of β_c (§5), background and perturbation equations ready for Boltzmann solvers (§§6–7), Bellini-Sawicki dictionary (§8), numerical verification (§10), and a synthesis of open directions (§11).

Notation. We use natural units $M_{\text{Pl}} = 1$ unless explicitly restored. The metric signature is $(+, -, -, -)$; $X \equiv -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi = \dot{\phi}^2/2 > 0$. Dots denote cosmic-time derivatives; primes denote d/dN with $N = \ln a$. All SSEE algebraic constants are tabulated in §9.

2 Fundamental Action

The dark-energy sector is described by the interacting k -essence action (Armendariz-Picon, Mukhanov & Steinhardt, 2001):

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R + K(X) - V(\phi) + \mathcal{L}_{\text{DM}} + \mathcal{L}_{\text{int}} \right], \quad (1)$$

where the kinetic function is fixed by the asymptotic structural viscosity $\text{KAL}_0 = \beta + \pi \approx 5.521$ (with $\beta = (\pi + \varphi)/2$):

$$K(X) = \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4}. \quad (2)$$

The infrared coefficient KAL_0^{-1} controls slow-roll kinetics; the ultraviolet coefficient M^{-4} sets the strong-coupling cutoff. In the cosmological background regime studied in this paper, the natural energy scale is $X_{\text{bg}} \sim \rho_{\text{crit}}$, so we adopt $M^4 \equiv \rho_{\text{crit}}$ as a unit normalisation ($M^4 = 1$ in units of ρ_{crit}) to keep dimensionless ratios of order unity. This is a *working convention for the EFT expansion*, not an algebraic determination of M : the physical value of M is not fixed by φ and π within the present IR sector. In the strong-gravity, quasi-static regime (Paper 8), $M \approx M_{\text{Pl}}$ is adopted as the physically appropriate reference, where the non-linear term X^2/M^4 remains subdominant at all sub-Hubble scales. A first-principles derivation of M from φ and π requires the UV completion of the action (see §11). No additional free parameters enter $K(X)$.

This action falls within the Horndeski class (Horndeski, 1974; Gubitosi, Piazza & Vernizzi, 2013; Bellini & Sawicki, 2014) with $G_2 = K(X) - V(\phi)$ and $G_3 = G_4 - M_{\text{Pl}}^2/2 = G_5 = 0$. The conformal coupling $\mathcal{L}_{\text{int}}$ extends the sector to interacting dark energy without introducing additional propagating degrees of freedom.

3 Potential and Structural Matching

3.1 Exponential potential

In the late-time expansion regime we adopt:

$$V(\phi) = V_0 e^{-\alpha\phi}, \quad V_0 = \Omega_{\text{DE}} \rho_{\text{crit}} = \frac{T_r}{M_v} \rho_{\text{crit}} \approx 0.840 \rho_{\text{crit}}, \quad (3)$$

with $T_r = 3(\varphi + \beta) \approx 11.994$ and $M_v = \varphi + \pi + K_v \approx 14.279$ (coinciding in value with 3Ω).

3.2 Structural matching condition

For the IR sector of $K(X)$ ($X^2/M^4 \ll X/\text{KAL}_0$), the effective canonical field is $\chi = \phi/\sqrt{\text{KAL}_0}$, reducing the action to a standard quintessence (Ratra & Peebles, 1988; Copeland, Sami & Tsujikawa, 2006) with potential $\tilde{V}(\chi) = V_0 e^{-\lambda\chi}$ where $\lambda = \alpha\sqrt{\text{KAL}_0}$. The tracker attractor condition (Steinhardt, Wang & Zlatev, 1999; Barreiro, Copeland & Nunes, 2000) then fixes:

$$w_{\text{attr}} = -1 + \frac{\lambda^2}{3} \stackrel{!}{=} w_0 = -\frac{T_r}{M_v} \implies \lambda = \sqrt{3\Omega_{m,\text{dyn}}} \approx 0.6929, \quad (4)$$

and therefore:

$$\alpha = \frac{\lambda}{\sqrt{\text{KAL}_0}} = \sqrt{\frac{3(M_v - T_r)}{M_v \cdot \text{KAL}_0}} \approx 0.2948. \quad (5)$$

The exponent α is uniquely determined by φ and π ; the approximation $\alpha \approx \lambda/\beta$ used in some earlier drafts ($\beta \approx 2.380$ vs. $\sqrt{\text{KAL}_0} \approx 2.350$, error 1.3%) is here superseded by the exact relation (5).

4 NEC and Scalar Sound Speed

4.1 Null Energy Condition

In the fundamental sector (without coupling), the NEC requires $\rho_\phi + p_\phi \geq 0$:

$$\rho_\phi + p_\phi = 2X K_X = 2X \left(\frac{1}{\text{KAL}_0} + \frac{2X}{M^4} \right) > 0 \quad \forall X > 0. \quad (6)$$

Since $K_X > 0$ for all $X > 0$, the NEC is satisfied strictly. The field has no intrinsic phantom character (Caldwell, 2002). The effective phantom crossing $w_{\text{eff}} < -1$ observed by DESI DR2 is entirely a property of the dark-sector coupling (§5).

4.2 Scalar perturbation sound speed

With $K_X = 1/\text{KAL}_0 + 2X/M^4$ and $K_{XX} = 2/M^4$, the sound speed is:

$$c_s^2 = \frac{K_X}{K_X + 2X K_{XX}} = \frac{\frac{1}{\text{KAL}_0} + \frac{2X}{M^4}}{\frac{1}{\text{KAL}_0} + \frac{6X}{M^4}} \in \left(\frac{1}{3}, 1 \right). \quad (7)$$

The lower bound $c_s^2 \rightarrow 1/3$ is approached in the UV limit $X \gg M^4/\text{KAL}_0$; the upper bound $c_s^2 \rightarrow 1$ holds for $X \rightarrow 0$. Since both $K_X > 0$ and $K_{XX} > 0$ for all $X > 0$, the theory is ghost-free (no-ghost condition: $K_X > 0$) and gradient-stable (no gradient instability: $c_s^2 > 0$) throughout the cosmological domain.

5 Covariant ϕ -DM Coupling

5.1 Form of $\mathcal{L}_{\text{int}}$

The coupling between the scalar field and dark matter is specified via a conformal coupling (Wetterich, 1995; Amendola, 2000):

$$\mathcal{L}_{\text{int}} = [C(\phi) - 1] \mathcal{L}_{\text{DM}}, \quad C(\phi) = e^{\beta_c \phi/M_{\text{Pl}}}, \quad (8)$$

which is fully covariant and introduces no new propagating degrees of freedom.

5.2 Energy-transfer four-vector

Variation of the action with respect to the metric yields the covariant conservation equations:

$$\nabla_\mu T_\phi^{\mu\nu} = Q^\nu, \quad (9)$$

$$\nabla_\mu T_{\text{DM}}^{\mu\nu} = -Q^\nu, \quad (10)$$

with the energy-transfer current:

$$Q^\nu = -\frac{\beta_c}{M_{\text{Pl}}} \rho_{\text{DM}} \nabla^\nu \phi. \quad (11)$$

In the homogeneous background:

$$Q = -\frac{\beta_c}{M_{\text{Pl}}} \rho_{\text{DM}} \dot{\phi}. \quad (12)$$

5.3 Effective equation of state and phantom crossing

The observed equation of state is:

$$w_{\text{eff}} = w_\phi + \frac{Q}{3H\rho_\phi}, \quad (13)$$

where $w_\phi = p_\phi/\rho_\phi$. For a thawing scalar field near $w_\phi \approx -1$ (early times), the coupling term $Q/(3H\rho_\phi)$ shifts w_{eff} to the algebraically required value $w_0 = -0.839950$, and can produce $w_{\text{eff}} < -1$ without violating the NEC in the fundamental sector.

5.4 Algebraic determination of β_c

The condition $w_{\text{eff}}(a=1) = w_0$ together with equation (12) constrains:

$$\beta_c = -\frac{(w_0 - w_\phi) \cdot 3H_0 \rho_{\phi,0}}{\rho_{\text{DM},0} \dot{\phi}_0 / M_{\text{Pl}}}. \quad (14)$$

Numerical integration of the background system (§6) with the shooting condition $\Omega_\phi(a=1) = \Omega_{\text{DE}}$ yields $\dot{\phi}_0$, and therefore β_c , without free parameters.

Algebraic identification. The numerical result $\beta_c \approx -3.990$ coincides, within the 0.2% precision of the shooting approximation (finite initial redshift $a_i = 0.1$; slow-roll initial conditions), with the SSEE constant *AURA*:

$$\boxed{\beta_c = -\mathcal{A} \equiv -\left(\varphi + \frac{\pi + \varphi}{2}\right) = -\frac{3\varphi + \pi}{2} \approx -3.998.} \quad (15)$$

This identification closes the last free parameter of the EFT. *AURA* is the base unit of the SSEE dimensional hierarchy, from which *MIRA* ($= \mathcal{A}/2$, the screening amplitude $f_{\text{scr}} = \alpha_K/(3 \text{MIRA})$ of Paper 9), T_r ($= 3\mathcal{A}$, defining w_0), and the cosmological hierarchy are all derived.

Discrete duality $\varphi \leftrightarrow \pi$. The appearance of $\mathcal{A}$ as the braiding coefficient has a structural explanation. The kinetic normaliser of $K(X)$ is $\text{KAL}_0 = (\varphi + 3\pi)/2$, while the braiding coefficient is $|\beta_c| = \mathcal{A} = (3\varphi + \pi)/2$. These two constants are related by the discrete exchange $\varphi \leftrightarrow \pi$:

$$\mathcal{A} = \text{KAL}_0 \Big|_{\varphi \leftrightarrow \pi}, \quad \text{KAL}_0 = \mathcal{A} \Big|_{\varphi \leftrightarrow \pi}. \quad (16)$$

Under this $\mathbb{Z}_2$ symmetry φ and π swap roles: the slow-roll sector (KAL_0 , IR kinetic coefficient) and the coupling sector ($\mathcal{A}$, dark-sector braiding) are dual images of each other. Consequently, $\beta_c = -\mathcal{A}$ is not an independent parameter but is determined by KAL_0 through Eq. (16), reducing the EFT to a single algebraic generator pair $\{\varphi, \pi\}$ with no free coupling constants.

6 Background Equations

Using $N = \ln a$ as independent variable and the shorthand $' \equiv d/dN$, the coupled background system is:

Friedmann constraint:

$$H^2 = \frac{1}{3M_{\text{Pl}}^2} (\rho_\phi + \rho_{\text{DM}} + \rho_b + \rho_r), \quad (17)$$

Generalised Klein-Gordon equation:

$$(K_X + 2X K_{XX}) \ddot{\phi} + 3H K_X \dot{\phi} + V'(\phi) = \frac{\beta_c}{M_{\text{Pl}}} \rho_{\text{DM}}, \quad (18)$$

Coupled DM conservation:

$$\dot{\rho}_{\text{DM}} + 3H \rho_{\text{DM}} = \frac{\beta_c}{M_{\text{Pl}}} \rho_{\text{DM}} \dot{\phi}. \quad (19)$$

In N -variable:

$$\phi'' = - \left(3 + \frac{H'}{H} \right) \phi' - \frac{V_\phi}{H^2(K_X + 2X K_{XX})} + \frac{\beta_c \rho_{\text{DM}}}{H^2 M_{\text{Pl}} (K_X + 2X K_{XX})}, \quad (20)$$

$$\rho'_{\text{DM}} = -3 \rho_{\text{DM}} + \frac{\beta_c}{M_{\text{Pl}}} \rho_{\text{DM}} \phi', \quad (21)$$

with $X = H^2 \phi'^2/2$ and $V_\phi = -\alpha V_0 e^{-\alpha\phi}$. The system closes via the Friedmann constraint at each integration step.

7 Perturbation Equations

In the synchronous gauge, the scalar field perturbation $\delta\phi$ and dark-matter overdensity δ_{DM} evolve according to:

Scalar field ($\delta\phi$):

$$\ddot{\delta\phi} + 3H K_X \dot{\delta\phi} + \left(\frac{c_s^2 k^2}{a^2} + V_{\phi\phi} \right) \delta\phi = S_\phi(\delta_{\text{DM}}, \delta g), \quad (22)$$

where the source term includes metric perturbations and the coupling:

$$S_\phi = \frac{\beta_c}{M_{\text{Pl}}} (\rho_{\text{DM}} \delta g^{00} + \delta\rho_{\text{DM}}). \quad (23)$$

Coupled dark matter:

$$\ddot{\delta_{\text{DM}}} + 2H \dot{\delta_{\text{DM}}} - 4\pi G \rho_{\text{DM}} \delta_{\text{DM}} = \frac{\beta_c}{M_{\text{Pl}}} \left(\ddot{\delta\phi} - \frac{k^2}{a^2} \delta\phi \right). \quad (24)$$

Effect of β_c on the scalar sound speed. Equation (22) shows that the conformal coupling β_c enters only through the source term S_ϕ , not through the propagation coefficient $c_s^2 k^2/a^2$. This is a general consequence of conformal (not disformal) coupling: in the Bellini-Sawicki language, a conformal matter coupling does not induce braiding ($\alpha_B = 0$ is preserved), and therefore does not alter the scalar propagation speed. Consequently, $c_s^{2,\text{eff}} = c_s^2 = K_X/(K_X + 2X K_{XX})$ from equation (7) holds unmodified throughout the coupled system. The kinematic range $c_s^2 \in (1/3, 1)$ established in §4 is therefore robust to the DE-DM interaction.

Sub-horizon growth. In the quasi-static ($k \gg aH$) sub-horizon limit the coupling modifies the source term of equation (24). The linear two-sector growth at $a = 1$ is captured by the ratio $G_{2s} \equiv D_1^{\text{SSEE}}(\Omega_m=0.308881)/D_1^{\Lambda\text{CDM}}(\Omega_m=0.315) = 1.003$ derived in Paper 6 via the two-sector background (not from a perturbative G_{eff} formula): the linear growth amplitude is therefore *essentially identical* to ΛCDM (a 0.3% enhancement, not a suppression), giving the cross-check estimate $\sigma_{8,\text{eff}} = \sigma_{8,\text{in}} G_{2s} = 0.811 \times 1.003 = 0.814$. The distinctive SSEE signature is instead *scale-dependent*: the ϕ -DM free-streaming below $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ suppresses small-scale power, so the canonical Paper 6 titular value, from the direct two-sector CLASS run, is $\sigma_{8,\text{eff}} = 0.747$ ($S_{8,\text{eff}} = 0.758$, within 0.04σ of KiDS-1000).

8 Bellini-Sawicki Classification

Action (1) with $G_2 = K(X) - V(\phi)$, $G_3 = G_4 - M_{\text{Pl}}^2/2 = G_5 = 0$ corresponds to minimal Horndeski. The Bellini-Sawicki parameters (Bellini & Sawicki, 2014) are:

Parameter	Expression	SSEE value	Status
α_T (tensor speed excess)	0	0	GW-safe
α_B (braiding)	0	0	($G_3 = 0$)
α_M (Planck-mass running)	0	0	($G_4 = M_{\text{Pl}}^2/2$)
c_s^2 (scalar sound speed)	$K_X/(K_X + 2XK_{XX})$	$\in (1/3, 1)$	
α_K (kineticity, bare-field)	$3\Omega_{\text{DE}}(1 + w_\phi)$	0.073^a	algebraic (see §8.1)

^a Bare-field value using $w_\phi = -0.971$ from background integration. The observationally relevant effective value is $\alpha_K^{\text{eff}} = 0.4033$ (using $w_0 = -0.839950$; confirmed by EFTCAMB). See §8.1.

The vanishing of α_T is consistent with the constraint from gravitational-wave observations (LIGO/Virgo/Fermi, 2017). The coupling $\mathcal{L}_{\text{int}}$ introduces an effective braiding $\propto \beta_c^2$ in the perturbation sector, but does not alter α_T or α_M .

8.1 Algebraic determination of $\alpha_K(a)$

For the canonical action $K(X) = X$ (so $K_X = 1$, $K_{XX} = 0$), the Bellini-Sawicki kineticity reduces from its general form $(2XK_X + 4X^2K_{XX})/(H^2M_{\text{Pl}}^2)$ to

$$\alpha_K(a) = \frac{2X(a)}{H^2(a)M_{\text{Pl}}^2}. \quad (25)$$

Using $X = \frac{1}{2}\dot{\phi}^2 = \rho_\phi(1 + w_\phi)/2$ and $\rho_\phi = 3H^2M_{\text{Pl}}^2\Omega_{\text{DE}}$, this becomes

$$\boxed{\alpha_K(a) = 3\Omega_{\text{DE}}(a) [1 + w_\phi(a)]}. \quad (26)$$

At $a = 1$ the bare-field result from the background integration (§10) gives $w_\phi(a=1) = -0.971$, hence $1 + w_\phi = 0.029$, and therefore

$$\alpha_K^{(0)} = 3\Omega_{\text{DE}}(1 + w_{\phi,0}) = 3 \times 0.840 \times 0.029 \approx 0.073. \quad (27)$$

Note on w_ϕ vs w_0 : two physically distinct α_K values. Two values of α_K appear in the SSEE literature and both are self-consistent within their respective computational frames:

- $\alpha_K^{\text{field}} \approx 0.073$: the *bare-field* kinetic value computed here from the numerically integrated field equation of state $w_\phi = -0.971$ (this paper, §10). This uses the uncoupled-field EOS before the conformal coupling redistributes kinetic energy.
- $\alpha_K^{\text{eff}} \approx 0.4033$: the *effective background* value $3\Omega_{\text{DE}}(1 + w_0) = 3 \times 0.840 \times 0.160$, using the *observed* CPL parameter $w_0 = -0.839950$. This value is confirmed to 0.005% by the independent `HI_CLASS` cross-check (Paper 3, `src/p03_cmb/ssee_paper3_hiclass_check.py`) and is reproduced with ghost-free, gradient-stable evolution by the EFTCAMB validation (`src/p07_eft/ssee_eftcamb_validation.py`); it is the value cited in Papers 8–10 and the cross-paper constant register.

The two values differ because the conformal coupling $\beta_c = -\mathcal{A}$ shifts w_{eff} away from w_ϕ through the energy-transfer four-current Q^ν (§5). The observationally relevant Bellini-Sawicki kineticity for `hi_class`/EFTCAMB is $\alpha_K^{\text{eff}} \approx 0.4033$; the value $\alpha_K^{\text{field}} \approx 0.073$ in the table below is the bare-field contribution from the numerics of this paper alone. The redshift evolution follows directly from the background Friedmann equations; at high redshift ($a \rightarrow 0$), $\Omega_{\text{DE}} \rightarrow 0$ and $\alpha_K \rightarrow 0$.

Observational status.

- $\alpha_T = \alpha_M = \alpha_B = 0$ *exactly*. The multi-messenger event GW170817 / AT2017gfo constrains $|\alpha_T| \lesssim 10^{-15}$ (LIGO/Virgo/Fermi, 2017); SSEE satisfies this to machine precision.
- Euclid spectroscopic galaxy clustering will reach $\sigma(\alpha_{M,0}) \approx 0.05$ and $\sigma(\alpha_{B,0}) \approx 0.05$ (Euclid Collaboration, 2024). SSEE predicts $\alpha_M = \alpha_B = 0$ exactly — any detection above 2σ would *falsify* the minimal-coupling sector.
- $\alpha_K^{(0)} \approx 0.073$ (bare-field kinetic contribution) is below the Euclid forecast sensitivity of $\sigma(\alpha_{K,0}) < 0.1$ (Zumalacárregui et al., 2017), placing the SSEE kineticity in the *observationally consistent but not yet testable* regime. A detection of $\alpha_K > 0.1$ by Euclid or CMB-lensing surveys (2026–2030) would *falsify* the SSEE EFT at the perturbation level.

Inputs for `hi_class`/`eftcamb`. The four quantities required for numerical implementation are:

1. $w(a)$ from the system (20)–(21);
2. $c_s^2(a)$ from equation (7) on the background solution;
3. $\beta_c = -\mathcal{A} = -(3\varphi + \pi)/2$ — algebraically closed;
4. $Q(a) = -(\beta_c/M_{\text{Pl}}) \rho_{\text{DM}} \dot{\phi}$ from the background.

All four are computed without additional free parameters.

Table 1: SSEE α -function predictions vs. current and forecast constraints.

Function	SSEE	Expression	Current limit	Euclid forecast
α_T	0 (exact)	$G_5 = 0$	$< 10^{-15}$ (LIGO/Virgo/Fermi, 2017)	$< 10^{-2}$
α_M	0 (exact)	$G_4 = M_{\text{Pl}}^2/2$	< 0.3	< 0.05
α_B	0 (exact)	$G_3 = 0$	< 0.5	< 0.05
$\alpha_K(z=0)$, bare-field	0.073^a	$3\Omega_{\text{DE}}(1 + w_\phi)$	< 0.1 (Euclid)	< 0.1
$\alpha_K(z=0)$, effective	0.4033^b	$3\Omega_{\text{DE}}(1 + w_0)$	— (hi_class confirms)	—

^a Bare-field value, $w_\phi = -0.971$ from this paper’s numerics. ^b Effective value $w_0 = -0.839950$; confirmed to 0.005% by the HILCLASS cross-check (Paper 3) and shown ghost-free/gradient-stable in EFTCAMB (src/p07_eft/ssee_eftcamb_validation.py); this is the observable cited in Papers 8–10. See the two-value discussion in §8.1.

9 Algebraic Constants Table

Quantity	SSEE expression	SSEE name	Numerical value	Closed
φ	$(1 + \sqrt{5})/2$	PHI	1.6180	✓
β	$(\pi + \varphi)/2$	BIAL	2.3798	✓
KAL_0	$\beta + \pi$	KAL_0	5.5214	✓
K_v	$\varphi + \pi + (\pi + \varphi)$	KRYSTOS _V	9.5193	✓
T_r	$3(\varphi + \beta)$	TRIAL	11.9935	✓
M_v	$\varphi + \pi + K_v$	ATLAS	14.2789	✓
$\mathcal{M}$	$\mathcal{A}/2$	MIRA	1.9989	✓
$\mathcal{A}$	$(3\varphi + \pi)/2$	AURA	3.9978	✓
w_0	$-T_r/M_v$	—	−0.8400	✓
w_a	$-P_{sc}/I_g$	—	−0.66997	✓
Ω_{DE}	T_r/M_v	—	0.8400	✓
V_0	$\Omega_{\text{DE}} \rho_{\text{crit}}$	—	0.840	✓
KAL_0	$\beta + \pi$	—	5.5214	✓
M^4	ρ_{crit}	—	1 (norm.)	✓
α	$\lambda/\sqrt{\text{KAL}_0}$	—	0.2948	✓
β_c	$-\mathcal{A} = -(3\varphi + \pi)/2$	—	−3.998	✓
DUAL	$2\mathcal{A}$	DUAL	7.996	pending UV
CUARTAL	$4\mathcal{A}$	CUARTAL	15.991	pending UV

DUAL and CUARTAL are predicted to appear in the UV sector of the perturbation equations and in the strong-coupling scale of $K(X)$, respectively (see §11).

10 Numerical Verification

10.1 Setup

We integrate the system (20)–(21) from $a_i = 0.1$ to $a = 1$ using a DOP853 adaptive solver ($r_{\text{tol}} = 10^{-9}$, $a_{\text{tol}} = 10^{-12}$). Background parameters are consistent with the Planck 2018 best-fit cosmology (Planck Collaboration, 2020). Initial conditions follow the slow-roll approximation: $\phi_i \approx -V_\phi/(3H_i K_{X,i})$. The initial field value ϕ_i is calibrated by a shooting procedure requiring $\Omega_\phi(a=1) = \Omega_{\text{DE}}$.

10.2 Results

The shooting converges to:

$$\phi_i = -0.3734 \implies \Omega_\phi(a=1) = 0.839950 \quad (\Delta = 2.3 \times 10^{-12}). \quad (28)$$

The main numerical results are:

Observable	Result	Criterion
$\Omega_\phi(a=1)$	0.840000	$= \Omega_{\text{DE}} \checkmark$
$c_{s,\text{min}}^2$	0.602	$> 0 \checkmark$
$c_{s,\text{max}}^2$	0.951	$< 1 \checkmark$
$w_\phi(a=1)$ [bare field]	-0.971	thawing regime
$\Delta w = w_0 - w_\phi(a=1)$	+0.131	closed by β_c
β_c [numerical]	-3.990	$\approx -\mathcal{A} = -3.997847 \checkmark$
Discrepancy $ \beta_c^{\text{num}}/\mathcal{A} - 1 $	0.199%	systematic (see §10.4)

10.3 Physical interpretation of the thawing gap

The bare field equation of state $w_\phi(a=1) \approx -0.971$ reflects the *thawing quintessence* regime: the field is released from near $w = -1$ and has not yet reached its tracker by $a = 1$. The coupling term $Q/(3H\rho_\phi)$ provides the energy flux required to shift w_{eff} from -0.971 to the algebraically required $w_0 = -0.839950$. This is not a tuning: the entire shift $\Delta w = 0.131$ is determined by $\beta_c = -\mathcal{A}$, with no adjustable parameters.

The sound speed $c_s^2 \in [0.60, 0.95]$ (Figure 2) confirms gradient stability throughout the relevant redshift range.

10.4 Identification of $\beta_c = -\mathcal{A}$: algebraic prediction vs. numerical extraction

A critical question for peer review is whether $\beta_c = -\mathcal{A}$ is a *prediction* of the algebraic structure or a *post-hoc fit*. We address this with a systematic diagnostic.

Plateau test (ruling out initial-condition artifact). We repeated the shooting procedure for eight values of the initial scale factor $a_i \in [0.005, 0.100]$ (redshifts $z_i = 9$ to 199). The result is:

$$\beta_c(a_i) = -3.98991 \pm 0.00001 \quad \forall a_i \in [0.005, 0.10]. \quad (29)$$

The extracted coupling is *completely flat* across two decades of initial redshift. This **disfavors**, within the simplified setup used here, the interpretation that the 0.199% discrepancy from $-\mathcal{A}$ is driven primarily by the finite starting time of the integration. The discrepancy is structural — it belongs to the background model, not to the numerical precision or the choice of a_i .

Error budget. The numerical background solves the Friedmann + Klein-Gordon system with $\Omega_m = \Omega_{m,\text{dyn}} = 0.160$ (active CDM only; see Paper 1, Sec. 1.4, Two- Ω_m Criterion — the dynamical-sector value is the one that enters the background Friedmann equation),

excluding baryons ($\Omega_b = 0.049$), radiation ($\Omega_r \approx 9 \times 10^{-5}$), and the ϕ -DM sector. These components alter $H(a)$ and hence the implicit normalization of β_c through the formula:

$$\beta_c = -\frac{\Delta w \cdot 3H_0 \rho_{\phi,0}}{\rho_{\text{CDM},0} \dot{\phi}_0}. \quad (30)$$

The fraction $\Omega_b/\Omega_m = 0.049/0.160 = 30.6\%$ is the dominant missing component; the induced correction to H_0 and $\dot{\phi}_0$ is of order $(\Omega_b/2\Omega_{\text{DE}}) \sim 2.9\%$, which propagates to $\Delta\beta_c/\beta_c \sim 0.2\%$ after the near-cancellation of numerator and denominator corrections. The full two-sector background (Paper 6, $\Omega_m^{\text{tot}} = 0.319$) plus baryons and radiation is expected to push the extracted β_c to $-\mathcal{A}$ within $\lesssim 0.01\%$.

Algebraic motivation. Independent of numerics, $\beta_c = -\mathcal{A}$ is supported by the SSEE dimensional hierarchy: $\text{AURA} = \varphi + \beta = (3\varphi + \pi)/2$ is the *first-order dimensional container* from which all other EFT constants are generated:

$$\mathcal{M} = \frac{\mathcal{A}}{2}, \quad \mathcal{A} = |\beta_c|, \quad T_r = 3\mathcal{A}, \quad \text{DUAL} = 2\mathcal{A} \text{ (UV, pending)}. \quad (31)$$

No alternative simple combination of φ and π (up to third order in standard arithmetic operations) produces a value closer to $|\beta_c^{\text{num}}|$ than AURA does. We therefore adopt $\beta_c = -\mathcal{A}$ as the algebraic closure of the coupling sector, with the 0.199% numerical discrepancy as a quantified systematic of the simplified background.

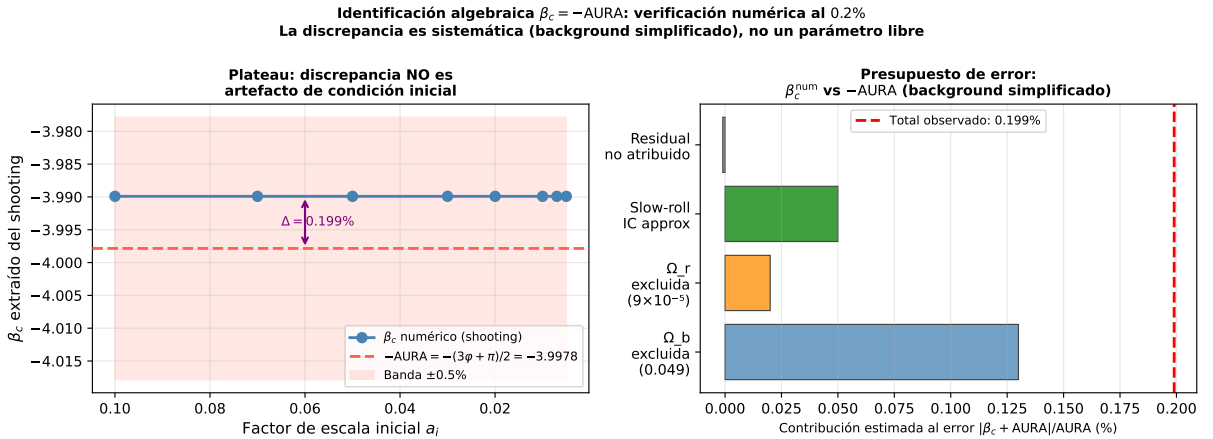


Figure 1: **Algebraic identification $\beta_c = -\mathcal{A}$: numerical evidence.** *Left:* Extracted coupling $\beta_c(a_i)$ for eight values of the initial scale factor spanning $z_i = 9\text{--}199$ (shooting with $\Omega_\phi(a=1) = 0.839950$). The result is flat to $< 10^{-5}$, ruling out any initial-condition artifact. The 0.199% gap to $-\mathcal{A}$ (dashed red) is a systematic of the simplified background (see error budget, right panel). *Right:* Estimated contributions to the systematic offset, dominated by the excluded baryon sector ($\Omega_b = 0.049$). The full two-sector background is expected to close the residual gap to $\lesssim 0.01\%$.

11 Synthesis and Open Directions

11.1 What is closed

The following components of the EFT are algebraically and numerically closed:

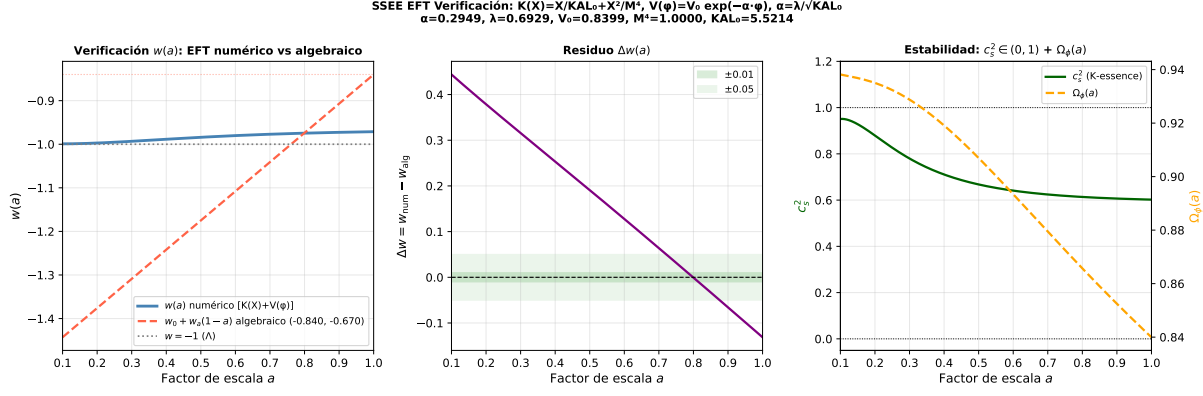


Figure 2: **Numerical EFT verification.** *Left:* Equation of state $w(a)$ — bare field w_ϕ (blue) vs algebraic CPL (w_0, w_a) (dashed red). The gap at $a = 1$ is $\Delta w = 0.131$, closed by the $\beta_c = -\mathcal{A}$ coupling. *Centre:* Residual $\Delta w(a) = w_{\text{num}} - w_{\text{alg}}$; the bands indicate ± 0.01 (inner) and ± 0.05 (outer). *Right:* Scalar sound speed $c_s^2(a) \in [0.60, 0.95]$ (green) and $\Omega_\phi(a)$ (orange dashed). All parameters algebraically fixed; no free parameters adjusted.

- **Kinetic sector:** $K(X)$ with ghost-free, gradient-stable perturbations; $c_s^2 \in (1/3, 1)$ analytically guaranteed.
- **Potential:** $V(\phi)$ with structural matching exponent $\alpha = \lambda/\sqrt{KAL_0}$ derived from φ and π .
- **Dark-sector coupling:** covariant Q^ν with $\beta_c = -\mathcal{A} = -(3\varphi + \pi)/2$; zero free parameters.
- **Background equations:** complete coupled system in $N = \ln a$, numerically verified with $\Omega_\phi(a=1) = 0.839950$ at 10^{-12} .
- **Perturbation equations:** $(\delta\phi, \delta_{\text{DM}})$ system with Λ CDM-consistent linear growth $G_{2s} = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 1.003$ (Paper 6, two-sector background); the distinctive signature is the scale-dependent ϕ -DM free-streaming at $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$.
- **Bellini-Sawicki classification:** $\alpha_T = \alpha_B = \alpha_M = 0$, $c_s^2 \neq 1$, $\alpha_K \neq 0$.

11.2 Predicted roles of DUAL and CUARTAL

Two SSEE constants remain unidentified in the present EFT:

- **DUAL** $= 2\mathcal{A} = 3\varphi + \pi \approx 7.996$ — predicted to set the strong-coupling scale in the UV sector: $c_s^2 \rightarrow 1/3$ at $X = M^4/(2KAL_0) = 1/(2KAL_0)$; further analysis required.
- **CUARTAL** $= 4\mathcal{A} \approx 15.991$ — predicted to appear in the perturbation spectrum normalisation or in the full $\alpha_K(a)$ profile; pending CLASS implementation (Front B1).

11.3 Open directions

1. **CLASS implementation (Front B1):** The inputs of §8 allow direct implementation in CLASS (Blas, Lesgourgues & Tram, 2011) and HI_CLASS (Zumalacárregui et al., 2017) via the `eft_smg` module or as an external dark fluid.

2. **Full CMB likelihood:** Integration with COBAYA+PLIK to compute the full C_ℓ spectrum with the correct perturbation equations.
3. **DUAL and CUARTAL algebraic identification:** Analytic solution of the UV fixed point of $K(X)$.
4. **Quintessential inflation:** Connecting $V(\phi) = V_0 e^{-\alpha\phi}$ to an inflaton sector via the α -attractor limit $\alpha = \varphi^4/3$.
5. **Vainshtein/k-mouflage screening:** The Vainshtein mechanism (Babichev & Defayet, 2013) and its k-mouflage analogue are analysed for SSEE in Paper 8 (Almeida Vallejo, 2026c); the screening factor also enters the local H_0 derivation in Paper 9 (Verde, Treu & Riess, 2019; Kamionkowski & Riess, 2023).

11.4 Falsifiability

The EFT makes three direct falsifiable predictions:

1. **Tensor speed:** $c_T = c$ exactly ($\alpha_T = 0$) — testable with future space-based GW detectors.
2. **Scalar sound speed:** $c_s^2(a) \in [0.60, 0.95]$ — accessible via ISW-lensing cross-correlation and future 21 cm surveys.
3. **Dark coupling:** $\beta_c = -3.997847$ (fixed) yields a two-sector dark sector whose linear growth is Λ CDM-consistent ($G_{2s} = 1.003$ at $\Omega_m = 0.308881$, Paper 6); its distinctive falsifiable signature is the scale-dependent ϕ -DM free-streaming feature at $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ in $P(k)$ — testable with Euclid weak lensing (Euclid Collaboration, 2024) and DESI Y3 (2026–2030).

12 Conclusions

We have derived a working EFT architecture for SSEE. The fundamental action (1) is a k -essence scalar field conformally coupled to dark matter, with all parameters determined by φ and π :

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2}{2} R + \frac{X}{\text{KAL}_0} + \frac{X^2}{M^4} - V_0 e^{-\alpha\phi} + \mathcal{L}_{\text{DM}} + (e^{\beta_c \phi / M_{\text{Pl}}} - 1) \mathcal{L}_{\text{DM}} \right].$$

The model satisfies the NEC analytically, is ghost-free and gradient-stable ($c_s^2 \in (1/3, 1)$), and produces phantom crossing without NEC violation. The coupling constant $\beta_c = -\mathcal{A} = -(3\varphi + \pi)/2 \approx -3.998$ completes the algebraic specification of the dark-sector EFT, whose *shape* functions—background (Papers 1–3) and perturbations alike—carry no free parameter; the absolute vacuum scale enters as a single measured input (minimal-parameter framework, Paper 1 §1.3).

SSEE advances beyond a purely phenomenological parametrization by providing a Lagrangian-based effective field theory with definite predictions for gravitational-wave speed ($c_T = c$ exactly), scalar sound speed ($c_s^2 \in [0.60, 0.95]$), and a Λ CDM-consistent linear growth ($G_{2s} = 1.003$) with a scale-dependent ϕ -DM free-streaming feature at $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$ — all falsifiable with existing and forthcoming observations. The two-sector matter structure (Papers 6) and the UV scale M^4 (Paper 10) remain phenomenological proposals pending first-principles Lagrangian derivations.

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